Day 4
Different datatypes in mysql

Here's a complete guide to the different data types in MySQL, grouped by category, along with their ranges and usage.

Numeric Data Type

Integer types

Data Type	Storage	Range (signed)	Range (unsigned)
tinyint	1 byte	-128 to 127	0 to 255
smallint	2 bytes	-32,768 to 32,767	0 to 65,535
mediumint	3 bytes	-83,88,608 to 83,88,607	0 to 1,67,77,215
int / integer	4 bytes	-2,14,74,83,648 to 2,14,74,83,647	0 to 4,29,49,67,295
bigint	8 bytes	-9 quintilion to +9 quintilion	0 to 18 quintilion

Decimal & Floating - point types

Data type	Description
decimal (p,s) or numeric(p,s)	fixed-point (precise) decimal numbers. p = precision, s = scale.
float(p)	Approximate (single-precision) floating-point number
DOUBLE / DOUBLE PRECISION	Approximate (double-precision) floating-point number

Date and Time types

Data type	Format	Range
date	'yyyy-mm-dd'	'1000-01-01' to '9999-12-31'
DATETIME	'YYYY-MM-DD HH:MM:SS'	'1000-01-01 00:00:00' to '9999-12-31 23:59:59'
TIMESTAMP	'YYYY-MM-DD HH:MM:SS'	'1970-01-01 00:00:01' UTC to '2038-01-19 03:14:07' UTC
TIME	'HH:MM:SS'	'-838:59:59' to '838:59:59'
YEAR	'YYYY'	1901 to 2155

String (Character)

Datatype	Description	Max length
CHAR(n)	Fixed-length string	0 to 255
VARCHAR(n)	Variable-length string	0 to 65,535
text	large text fields	65,535 bytes
tinytext	up to 255 MB	255
mediumtext	up to 16 MB	16,7,77,215
longtext	up to 4 GB	4,29,49,67,295
enum	A predefined list of allowed values	1 to 65,535 values
set	A set of values (can select multiple)	Up to 64 members

Binary Data Types

Datatype	Description	Max Length
binary(n)	Fixed-length binary data.	0 - 255
varbinary(n)	variable length binary data.	0 - 65,535
tinyblob	tiny binary object	255 bytes
blob	binary large object	65,535 bytes
mediumblob	Medium binary object	16 MB
longblob	Long binary object	4 GB

Boolean type

Datatype	Description
boolean or bool	internally stored as tinyint(1) (0 = false, 1= true)